

LUMBINI ENGINEERING COLLEGE (LEC)
INTERNAL EXAMINATION 2026

Level: Bachelors degree
Programmed: BE Computer 7th Sem.
Course: Engineering Economics

Full Marks: 100
Pass Marks: 40
Time: 3 hrs

Candidates are required to give their answer in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions:

1. a) What is utility? Explain about the law of diminishing returns with focusing its stages. 7
- b) What do you mean by time value of money? Express cash flow as uniform payment series for 10 years that is accumulated value of a payment series of Rs. 50,000 at the end of first year, decreasing by Rs. 3,000 each year for rest 9 years at interest rate of 16% per year compounded annually 8
2. a) What do you mean by IRR & ERR? Evaluate the following project with the help of ERR for external reinvestment rate 20% per year and MARR 16% per year. 7
- | | |
|--------------------|---------------|
| Initial investment | =Rs. 6,00,000 |
| Expected life | =8 Years |
| Salvage value | =Rs. 1,00,000 |
| Annual expenses | =Rs. 90,000 |
| Annual revenue | =Rs. 200,000 |
- b) Evaluate the project having following cash flow with both types of B/C ratio using by PW method. 8
- | | |
|------------------------|----------------|
| Initial investment | =Rs. 8,00,000 |
| Salvage value | =Rs. 150,000 |
| Annual revenue | =Rs. 190,000 |
| Annual expenses | =Rs. 50,000 |
| MARR | = 12% per year |
| Useful life of project | = 10 Years |
3. a) perform sensitivity analysis using PW Method over a range of (+ or -) 30% in
- | | | | |
|------|---------------------|-----|--------------------|
| i. | initial investment. | ii. | net annual revenue |
| iii. | salvage value | iv. | useful life: |
- | | |
|-------------------------|---------------|
| Initial investment (Rs) | = 200,000 |
| Annual revenue (Rs) | =60,000 |
| Annual expenses (Rs) | =10,000 |
| Salvage values (Rs) | =25,000 |
| Useful life | =10 yrs |
| MARR | =12% Per year |
- Draw also the sensitivity graph. 8
- b) What do you mean by Inflation? Briefly explain about the effects of Inflation. 7
4. a) Evaluate the following two feasible investments A and B having different useful life, if MARR is 15 % per year. Use PW method with repeatability assumptions. 8

Details	For Project A (Rs.)	For Project B (Rs.)
Investment	40,000	50,000
Net annual revenue	15,000	20,000
Salvage value	5,000	6,000
Useful life	3 years	5 years

b) Determine the choice between a defender that has a current market value of Rs. 5,000 & a Challenger that can be purchased for Rs. 7,500. Both have service life of 3 years with no salvage value at the end of life & MARR is 12%. Their operating costs are as below:-

Year	Defender (D)	Challenger (C)
1	1800	500
2	2200	800
3	2800	1200

5. a) What do you mean by Defender & Challenger? Describe about the reasons of replacement. 7

b) Find internal rate of return for the project whose cash flows are as follows: 8

Initial investment	= Rs. 5,00,000
Annual benefit	= Rs. 1,50,000
Annual cost	= Rs. 30,000
Salvage value	= Rs. 40,000
MARR	= 12% per year

6. a) Define depreciation. What are the causes for it? If a machine costing Rs. 1,50,000 is purchased by expecting salvage value of Rs. 40,000 at the end of 6th year. Calculate depreciation amount for each year by Sum of Year digit method. 7

b) Consider the following cash flow:

End of year	Net cash flow (Rs,000)
0	-1100
1	-550
2	0
3	230
4	1000
5	520
6	320
7	450

Calculate the present worth and future worth of that cashflow of the project. if MARR is 15% per year. Do this project economically feasible? 8

7. Write short notes on: (Any two)

- Life cycle of cost.
- Supply & Law of supply.
- Payback period.

2*5=10